

Task Tracker Application - Bug Fix

Bug #1 → Status Filtering not working correctly

* Where - 1) TaskRepository.java (due to parenthesis)
2) search-task.sql
3) task-search-package.sql } In db folder

* How I found it

- While testing the application, I changed the status from the dropdown like Open / InProgress / Done, but the task list was not changing correctly.
- Also tested the API using curl and noticed that archived tasks could appear in the results even though they should be hidden.

* Root cause

- SQL WHERE condition
- The query was using both AND and OR without proper brackets.
- In SQL, AND is evaluated before OR

*Before

```
WHERE archived = FALSE  
AND LOWER(title) LIKE :term  
OR LOWER(description) LIKE :term  
AND (:status IS NULL OR status = :status)
```

here

- here because of the operator precedence, the status and archived conditions were not being applied to all search results.
- That is why selecting a status could still show that task that should have been filtered out.

*After

- added brackets around the title / description search condition

```
WHERE archived = FALSE  
AND (  
  LOWER(title) LIKE :term  
  OR LOWER(description) LIKE :term  
)  
AND (:status IS NULL OR status = :status)
```

Why this fix?

- Now the search first checks the title OR description
- After that archived = FALSE and the selected status are applied to the matching tasks.

- This makes sure all three condⁿ work together.

Result:

- Tested diff status values again
- Open, In Progress and Done now return the correct tasks
- Archived tasks are also excluded from the search result.